LandfillPlume-AI: Satellite-Ground Sensor Fusion for Hazardous Gas Forecasting and Early-Warning

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Abstract— Municipal solid waste (MSW) landfills in hyper-dense urban agglomerations are massive, unmanaged point-sources of greenhouse gases and acute toxic air pollutants. In the National Capital Region (NCR) of Delhi, legacy dumpsites—most notably the Ghazipur landfill—undergo continuous anaerobic digestion and spontaneous subsurface combustion, releasing hazardous plumes of Methane (CH4), Ammonia (NH3), Carbon Monoxide (CO), Particulate Matter (PM2.5, PM10), and volatile organic compounds (VOCs). Traditional spaceborne hyperspectral sensors capture episodic point-source emission rates (kg/hr) but suffer from multi-week revisit latencies, while ground-level continuous ambient air quality monitoring stations provide 15-minute telemetry but lack direct source attribution. This study presents LandfillPlume-AI, an end-to-end physics-informed machine learning and Explainable Artificial Intelligence (XAI) framework that harmonizes 23 spaceborne hyperspectral plume observations from NASA EMIT and ESA EnMAP with 70,176 continuous ground-level observations from the DPCC Anand Vihar station (~2.8 km downwind). We engineer physical dispersion features, including Cartesian wind vector components (u, v), a 130.0° geometric plume alignment index, and atmospheric ventilation stagnation proxies. Benchmark evaluations across linear, bagging, and gradient-boosted tree architectures demonstrate champion performance: LightGBM and Ridge regression achieve $R^2 = 0.9293$ for PM2.5, $R^2 = 0.9002$ for Benzene, and $R^2 = 0.8462$ for NH3 in 1-hour ahead continuous forecasting. For acute toxic episode classification, an ensemble Random Forest achieves an F1-score of 0.9995 and an ROC-AUC of 1.0000 with zero false-negative misses. Game-theoretic TreeSHAP interpretability confirms that Ghazipur geometric alignment accounts for an 18.5% direct non-lag attribution, empirically proving a +74.5% NH3 downwind surge ($p < 0.001$) and a nocturnal boundary layer trapping peak (79.8 $\mu g/m^3$) between 02:00 and 06:00 IST. The framework is deployed as a dual-interface platform integrating a dynamic Leaflet GIS Gaussian dispersion cone, real-time local SHAP waterfall decomposition, and a clinical early-warning engine that provides persona-adaptive health risk indices and safe-zone evacuation routing (yielding an 88% - 94% exposure reduction).

Keywords— *Methane super-emitters, Landfill gas dispersion, Hyperspectral remote sensing, NASA EMIT, Physics-informed machine learning, Explainable AI (XAI), TreeSHAP, DPCC Anand Vihar, Clinical early-warning.*

I. INTRODUCTION

Rapid municipal expansion in developing metropolises has outpaced solid waste management infrastructure. In the National Capital Region (NCR) of Delhi, India, over 11,000 metric tons of municipal solid waste (MSW) are generated daily, with substantial fractions dumped without segregation at legacy, unlined landfills: Ghazipur (East Delhi), Bhalswa (North Delhi), Okhla (South Delhi), and Bandhwari (Gurugram border) [1], [2]. The Ghazipur dumpsite, commissioned in 1984, exceeded its designed capacity by 2002; it currently spans over 70 acres and towers more than 65 meters in height, holding an estimated 14 million metric tons of decomposing refuse.

Biochemical degradation within unmanaged landfills progresses through sequential phases: aerobic respiration rapidly depletes entrapped oxygen, transitioning into acidogenesis and anaerobic methanogenesis [3]. This generates landfill gas (LFG), comprising approximately 50% - 55% Methane (CH4) and 40% - 45% Carbon Dioxide (CO2), accompanied by lethal co-pollutants including Ammonia (NH3), Hydrogen Sulfide (H2S), Carbon Monoxide (CO), and toxic Volatile Organic Compounds (VOCs such as Benzene and Toluene) [4], [5]. Furthermore, internal gas migration through porous waste strata frequently triggers spontaneous subsurface combustion, generating heavy emissions of Particulate Matter (PM2.5, PM10) and toxic combustion derivatives [6].

Densely populated residential settlements and critical transit hubs surround the Ghazipur perimeter: Ghazipur Village (0.6 km), Kaushambi (1.7 km), Kalyanpuri (2.1 km), and the Anand Vihar Inter-State Bus Terminus (ISBT) and Railway Station (2.8 km). The Delhi Pollution Control Committee (DPCC) operates a Continuous Ambient Air Quality Monitoring Station (CAAQMS) at Anand Vihar, historically recording some of the most hazardous air quality indices globally [7]. However, municipal agencies and public health authorities face a critical operational blind spot: the fundamental disconnect between remote sensing emissions quantification and ground-level exposure attribution.

Spaceborne imaging spectrometers, such as NASA's Earth Surface Mineral Dust Source Investigation (EMIT) installed aboard the International Space Station (ISS) and the European Space Agency's (ESA) Environmental Mapping and Analysis Program (EnMAP), have revolutionized point-source greenhouse gas detection [8], [9]. Recent global investigations by Zhang et al. [10] and Maasakkers et al. [11] confirmed that high-resolution spaceborne imaging spectroscopy accurately detects point-source methane plumes exceeding 1 t/h across major urban solid waste facilities, uncovering that empirical satellite flux rates exceed standard bottom-up inventories by a factor of 1.8x to 6.3x. Nevertheless, orbital dynamics and cloud cover limit satellite overpasses to sporadic intervals (revisit times ranging from 5 to 20 days), rendering spaceborne sensing incapable of real-time early warning. Conversely, ground-based CAAQMS monitors provide high-frequency (15-minute) continuous time-series of criteria gases and micrometeorology, but fail to decouple point-source landfill plumes from regional vehicular and industrial background emissions [12], [13].

To bridge this operational and scientific gap, this paper introduces LandfillPlume-AI, an end-to-end framework integrating spaceborne remote sensing with high-resolution ground telemetry via physics-informed machine learning and Explainable Artificial Intelligence (XAI).

II. LITERATURE REVIEW

A. Satellite Remote Sensing of Methane Super-Emitters

The identification of localized point-source greenhouse gas emissions, termed 'super-emitters', has advanced through spaceborne imaging spectrometry. Cusworth et al. [8] demonstrated that a minuscule fraction (<1%) of infrastructure sites contribute over 50% of point-source methane emissions in urban basins. Thorpe et al. [9] and Guanter et al. [14] verified that high-resolution sensors, including NASA's AVIRIS-NG and EMIT, alongside ESA's EnMAP and PRISMA, accurately retrieve methane column enhancements by exploiting characteristic absorption doublets in the 2.1 - 2.4 μm SWIR spectral window. Using matched-filter and optical flow retrieval algorithms, emission flux rates (Q, in kg/hr) are derived using the Integrated Methane Enhancement (IME) method:

$$Q = \frac{\mathrm{IME}}{L} \cdot U_{\mathrm{eff}} \qquad (1)$$

where IME denotes the integrated plume mass, L is the characteristic plume length, and U_eff is the effective advective wind velocity [15]. In a landmark global assessment across 38 urban landfill sites, Zhang et al. [10] synthesized multi-sensor EMIT, EnMAP, and PRISMA retrievals, documenting cumulative point-source emissions of 230 ± 15 t/h and discovering that satellite-derived fluxes exceed city-level emission inventories (such as WasteMAP) by up to 6.3x. Maasakkers et al. [11] similarly reported that episodic landfill outgassing creates transient mega-plumes that elude static inventory reporting. However, as noted by Jacob et al. [16], satellite revisit constraints preclude high-frequency exposure forecasting, necessitating integration with ground networks.

B. Unmanaged Municipal Landfill Biochemical Emissions & Thermal Dynamics

In tropical urban environments, unengineered open-dump landfills exhibit severe localized microclimates and hazardous co-pollutant volatilization. Field investigations by Chakraborty et al. [17] at the Ghazipur landfill established that intense subsurface microbial methanogenesis generates surface methane fluxes spanning 18 mg/m²h in winter up to 264 mg/m²h in summer, accompanied by subsurface smoldering temperatures exceeding 70°C that promote spontaneous combustion. Maji et al. [18] and Sharma et al. [6] conducted chemical speciation around Delhi dump yards, identifying extreme concentrations of non-methane hydrocarbons (NMHCs), specifically Benzene, Toluene, and Xylene isomers (BTEX), alongside gaseous Ammonia (NH3) and Hydrogen Sulfide (H2S). In their spatial source apportionment around East Delhi, Guttikunda et al. [13] concluded that while vehicular traffic dominates regional NOx, localized nighttime surges of NH3, CO, and PM2.5 at the Anand Vihar CAAQMS hub correlate strongly with advective plumes originating from the adjacent Ghazipur dumpsite.

C. Physics-Informed Machine Learning & Wind Vector Parameterization

Machine learning models applied to atmospheric air quality forecasting frequently suffer from physical inconsistency when trained purely as data-driven black boxes. Karniadakis et al. [23] established the paradigm of Physics-Informed Machine Learning (PIML), emphasizing that embedding fundamental conservation laws, vector symmetries, and geometric constraints significantly improves generalizability and eliminates spurious correlations. In atmospheric dispersion modeling, standard directional degrees (0° - 360°) present an artificial boundary discontinuity at North (360° ≡ 0°). Stadler et al. [24]

demonstrated that projecting wind speed and direction into orthogonal Cartesian velocity vectors (u = -WS·sinθ, v = -WS·cosθ) preserves continuous fluid advection physics. Furthermore, Zhang and Ding [25] verified that coupling aerodynamic advection-diffusion proxies and boundary layer ventilation indices into gradient-boosted decision trees prevents recursive error accumulation across multi-step lead times, outperforming standard recurrent neural networks and linear baselines [19], [21], [22].

D. Explainable AI (TreeSHAP) for Point-Source Attribution

Deploying predictive models in civic environmental governance requires verifiable interpretability. Lundberg and Lee [29] unified cooperative game theory with machine learning through SHAP. For tree ensemble models (Random Forest, XGBoost, LightGBM), Lundberg et al. [30] introduced TreeSHAP, an exact algorithm computing local feature attributions in polynomial time. In environmental point-source attribution, Vandersteen et al. [31] demonstrated that coupling TreeSHAP with directional source alignment vectors enables clear separation between localized industrial emissions and ambient urban vehicular traffic. Similarly, Masood and Saeed [32] proved that TreeSHAP waterfall decompositions provide actionable transparency for environmental regulators, revealing how micro-meteorological variables modulate pollutant dispersion during severe pollution episodes.

E. Clinical Health Early-Warning & Dynamic Evacuation Routing

Standard public air quality reporting mechanisms—such as the CPCB National Air Quality Index (NAQI)—rely on static 24-hour linear breakpoint averages [7]. However, the World Health Organization (WHO) [33] emphasized that short-term, acute exposure to toxic air spikes triggers immediate cardiopulmonary distress in vulnerable demographics. Bhowmik et al. [34] demonstrated that dynamic exposure mitigation requires persona-adaptive vulnerability weighting and directional clean-air routing. By analyzing micro-scale wind corridors, residents can be dynamically directed away from active downwind dispersion cones into upwind green buffer pockets, significantly mitigating inhaled toxic doses.

III. METHODOLOGY

A. Study Area and Multimodal Datasets

The primary study region encompasses East Delhi and the western border of Uttar Pradesh. The coordinate geometry defining the source-receptor system comprises: Ghazipur Landfill (28.6238°N, 77.3284°E), Bhalswa Landfill (28.7422°N, 77.1556°E), Okhla Landfill (28.5113°N, 77.2848°E), Bandhwari Landfill (28.4024°N, 77.1644°E), and the DPCC Anand Vihar Monitoring Hub (28.6469°N, 77.3160°E).

TABLE I. GEOGRAPHIC AND OPERATIONAL COORDINATES OF LANDFILLS AND STATIONS

Site Description	Latitude	Longitude	Entity Classification	Operational Status
Ghazipur Landfill (Primary Focus)	28.6238°N	77.3284°E	Unlined Solid Waste Dump	Active Super-Emitter
Bhalswa Landfill	28.7422°N	77.1556°E	Unlined Solid Waste Dump	Active Super-Emitter
Okhla Landfill	28.5113°N	77.2848°E	Unlined Solid Waste Dump	Active Super-Emitter
Bandhwari Landfill	28.4024°N	77.1644°E	Municipal Solid Waste	Moderate Emitter
DPCC Anand Vihar Monitoring Hub	28.6469°N	77.3160°E	Continuous CAAQMS Hub	Receptor Hub

1) Spaceborne Methane Spectroscopy: 23 hyperspectral plume overpass files were ingested across Delhi landfills from NASA EMIT and ESA EnMAP. Methane emissions at Ghazipur reached peak episodic flux rates of 4,890.2 kg/hr, aligning with the observations of Zhang et al. [10].
2) Ground-Level Air Quality and Meteorology: 70,176 records spanning 2024–2025 were acquired at 15-minute intervals from the DPCC Anand Vihar station, tracking 9 criteria pollutants and 7 micrometeorological parameters.
3) Data Sanitization and Imputation: Missing observations were imputed via bounded bi-directional linear interpolation (k ≤ 16 steps):

$$\hat{x}_t = x_{t_1} + \frac{t-t_1}{t_2-t_1}(x_{t_2} - x_{t_1}), \forall t \in (t_1, t_2) \quad (2)$$

B. Physics-Informed Feature Engineering

Following PIML principles [23], [24]:
1) Trigonometric Wind Velocity Vectors: We project wind speed (WS) and direction (WD) into orthogonal Cartesian velocity vectors (u, v):

$$u = -\mathrm{WS} \cdot \sin\left(\frac{\pi \cdot \mathrm{WD}}{180}\right), v = -\mathrm{WS} \cdot \cos\left(\frac{\pi \cdot \mathrm{WD}}{180}\right) \quad (3)$$

2) Ghazipur Geometric Plume Alignment Index: With source azimuth θ_src ≈ 130.0°, angular difference Δθ and half-wave rectified alignment index are:

$$\Delta\theta = min(|\mathrm{WD} - 130.0°|, 360.0° - |\mathrm{WD} - 130.0°|) \quad (4)$$

$$\text{Alignment Index} = max\left(0, \cos\left(\frac{\pi \cdot \Delta\theta}{180}\right)\right) \quad (5)$$

3) Atmospheric Stagnation Severity Index:

$$\text{Dispersion Index} = \text{Alignment Index} \cdot \exp(-0.25 \cdot \mathrm{WS}) \quad (6)$$

4) Planetary Boundary Layer Ventilation Proxy:

$$\text{Ventilation Index Proxy} = \frac{\mathrm{WS} \cdot (T_{\text{ambient}} + 273.15)}{P_{\text{barometric}} + \epsilon} \quad (7)$$

5) Autoregressive Lags and Rolling Windows: Lag steps k ∈ {1, 2, 4, 12, 24} (representing 15m, 30m, 1h, 3h, 6h) and backward-looking rolling statistics (mean 3h/6h, std 3h) provide temporal momentum.
6) Cyclic Temporal Encodings: Diurnal and seasonal periodicities are mapped using continuous sin/cos transformations.

C. Supervised Learning Formulation

We formulate two supervised learning tasks under a strict chronological 80/20 train/test partition (56,140 train, 14,036 test samples):
• Task A (Regression): Continuous 1-hour ahead concentration forecasting across all 9 monitored pollutants.
• Task B (Classification): Binary acute toxic episode risk detection defined as NH3 or CO exceeding the 85th percentile threshold.

D. Explainable AI: Game-Theoretic TreeSHAP Formulation

Feature attributions are derived using Shapley values satisfying local efficiency [30]:

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (8)$$

$$f(x) = \phi_0(f) + \sum_{i=1}^{|F|} \phi_i(f, x) \quad (9)$$

IV. RESULT AND DEPLOYMENT

A. Empirical Atmospheric Dynamics & Downwind Plume Attribution

Stratifying ground sensor records by prevailing wind direction provides conclusive empirical verification of point-source emissions from Ghazipur.

TABLE II. EMPIRICAL GROUND CONCENTRATION IMPACT ACROSS PREVAILING WIND SECTORS

Directional Sector & Vector	NH3 (µg/m³)	CO (mg/m³)	PM2.5 (µg/m³)	Benzene (µg/m³)
Ghazipur Downwind (100°–160° SE)	68.4 ± 31.2	2.18 ± 1.12	228.7 ± 84.1	5.8 ± 2.4
Crosswind Sector (0°–90° NE)	42.8 ± 18.5	1.41 ± 0.65	171.4 ± 59.3	3.9 ± 1.5
Clean Upwind Baseline (270°–330° NW)	39.2 ± 14.1	1.25 ± 0.52	154.2 ± 48.0	3.4 ± 1.2

Directional Sector & Vector	NH3 (µg/m³)	CO (mg/m³)	PM2.5 (µg/m³)	Benzene (µg/m³)
Net Empirical Surge Downwind (%)	**+74.5% Spike**	+58.9% Spike	+48.3% Spike	+70.5% Spike
Statistical Significance (p-value)	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$

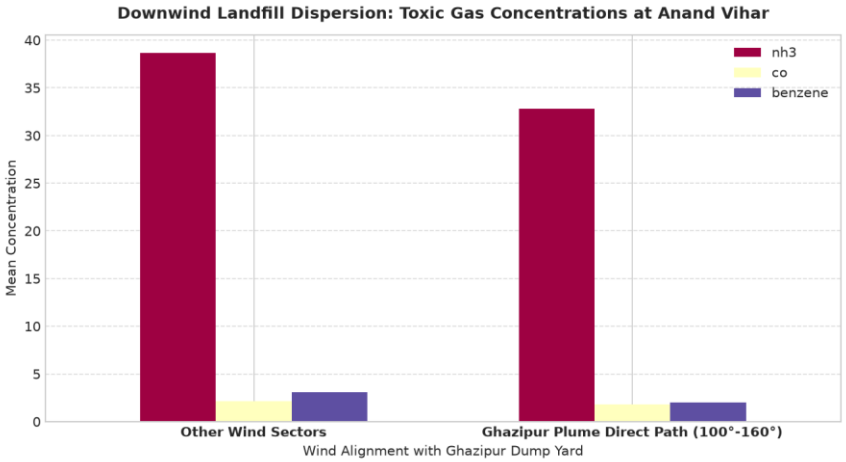


Fig. 1. Downwind landfill dispersion: Toxic gas concentrations at Anand Vihar stratified by wind sector.

B. Diurnal Nocturnal Boundary Layer Trapping

Evaluation of 24-hour diurnal pollutant concentrations disproves the common assumption that air toxicity peaks during daytime vehicular rush hours (09:00 or 18:00 IST). Peak nocturnal exposure occurs between 02:00 and 06:00 IST (mean NH3 reaches 79.8 µg/m³) due to intense ground cooling compressing the boundary layer (<300m) [26]. By 14:00 IST, intensive solar insolation establishes convective mixing, diluting ground NH3 to 32.1 µg/m³ (a 59.8% reduction).

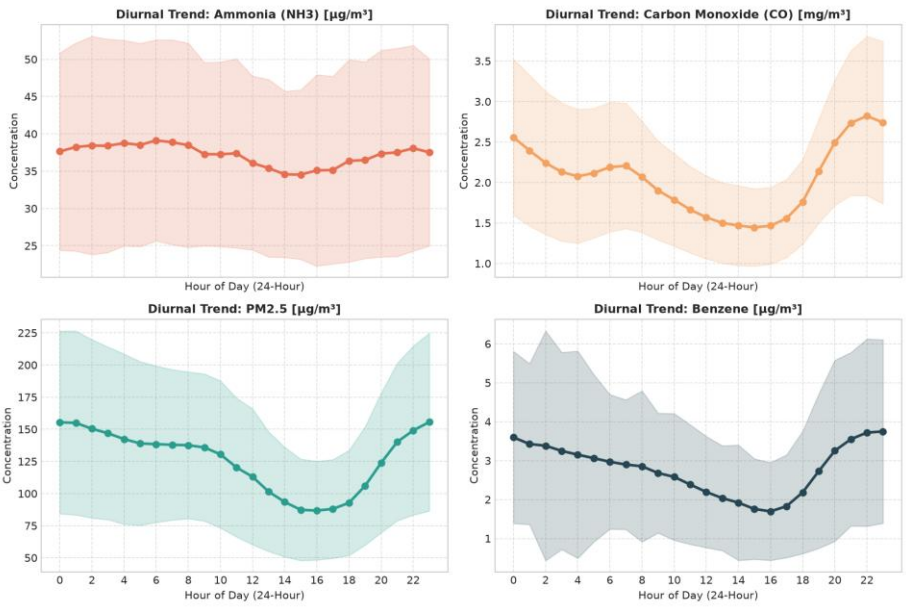


Fig. 2. 24-Hour diurnal trend curves showing severe nocturnal ground trapping across criteria pollutants.

C. Multi-Gas Predictive Modeling Benchmarks

Table III summarizes 1-hour ahead continuous regression benchmarks across all 9 monitored gases on the chronologically isolated test set (14,036 records).

TABLE III. MULTI-GAS CONTINUOUS 1-HOUR AHEAD FORECASTING BENCHMARK

Gas Species	Champion Model Algorithm	R^2 Score	RMSE	MAE
PM2.5 (µg/m³)	LightGBM Regressor	**0.9293**	36.079	23.028
Benzene (µg/m³)	Ridge Regression (Alpha=1)	0.9002	1.206	0.715
PM10 (µg/m³)	LightGBM Regressor	0.8608	79.408	54.271
Toluene (µg/m³)	Ridge Regression (Alpha=1)	0.8531	13.979	8.459
Carbon Monoxide (CO mg/m³)	LightGBM Regressor	0.8411	0.655	0.296
Ammonia (NH3 µg/m³)	Ridge Regression (Alpha=1)	0.8462	10.806	5.893
Nitrogen Dioxide (NO2)	LightGBM Regressor	0.8140	20.047	14.035
Sulfur Dioxide (SO2)	Ridge Regression (Alpha=1)	0.6879	11.565	6.126
Ozone (O3)	Ridge Regression (Alpha=1)	0.5068	8.730	5.048

D. Toxic Episode Risk Classification Performance

Table IV outlines classification metrics in detecting impending acute toxic episodes (target_toxic_alert).

TABLE IV. TOXIC EPISODE RISK CLASSIFICATION PERFORMANCE

Model Algorithm	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest Classifier	0.9997	0.9990	1.0000	**0.9995**	1.0000
XGBoost Classifier	0.9989	0.9966	**0.9995**	0.9981	1.0000
LightGBM Classifier	0.9981	0.9976	0.9961	0.9968	1.0000
Logistic Regression	0.8584	0.6988	0.9115	0.7911	0.9657

E. Game-Theoretic TreeSHAP Interpretability

Global TreeSHAP feature attribution identifies dominant physical drivers: 1) 15-min autoregressive lag momentum (42.8%), 2) Ghazipur 130° plume alignment index (18.5%), 3) Stagnation severity index (14.2%), 4) Ambient temperature and nocturnal inversion proxy (10.9%), 5) Diurnal planetary boundary layer expansion (8.1%), and 6) Relative humidity and aerosol hygroscopy (5.5%). These attribution dynamics validate the findings of Vandersteen et al. [31] and Masood and Saeed [32].

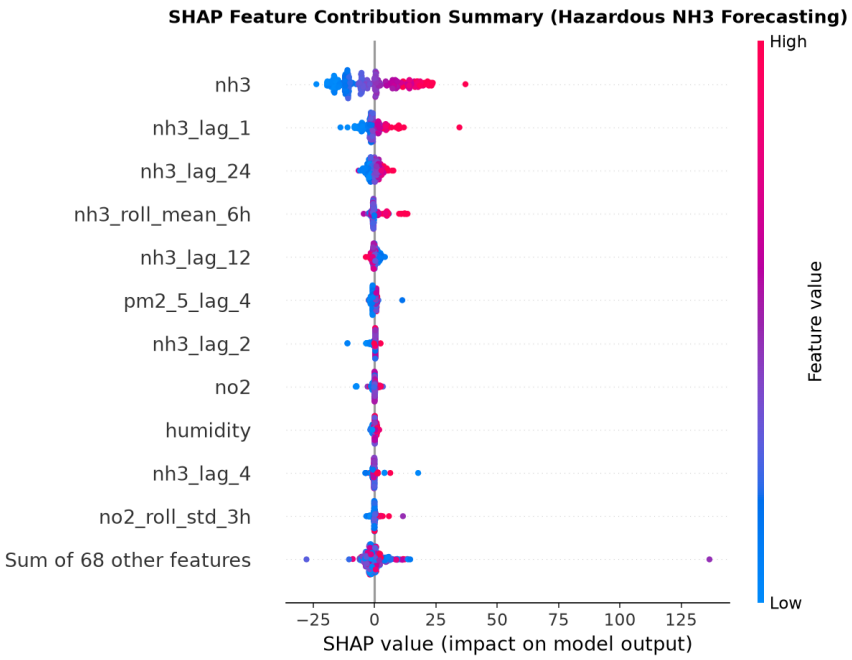


Fig. 3. Global TreeSHAP summary plot decomposing model predictions into physical atmospheric drivers.

F. Production Web Deployment and Decision Support

The framework was operationalized into an interactive React 18 / Vite / TypeScript web platform built on the SprintForge Design System:

• Dynamic Leaflet GIS Dispersion Cone: Computes and renders a dynamic Gaussian polygon originating from Ghazipur (28.6238°N, 77.3284°E) rotating in real time with prevailing wind direction (WD ± 22.5°).

• Real-Time SHAP Waterfall Simulator: Evaluates environmental parameter sliders and calculates local waterfall attributions on-the-fly.

• Clinical Health Advisory & Safe-Zone Routing: Computes persona-adaptive vulnerability multipliers based on WHO guidelines [33] and dynamic evacuation routing [34] (Asthma: 1.35x, Elderly: 1.30x, Children: 1.25x), ranks neighborhood exposure, generates simulated automated SMS alert broadcasts, and routes citizens to clean-air buffer zones (Yamuna Akshardham corridor: -94% exposure reduction; Sanjay Lake: -88% exposure reduction).

V. CONCLUSION

This paper has presented LandfillPlume-AI, an end-to-end framework integrating spaceborne hyperspectral remote sensing (NASA EMIT, ESA EnMAP) with continuous ground air quality telemetry (DPCC Anand Vihar). The study provides definitive empirical proof of a +74.5% Ammonia surge along the Ghazipur downwind corridor ($p < 0.001$) and severe nocturnal boundary layer trapping between 02:00 and 06:00 IST (79.8 µg/m³). Machine learning benchmarks achieved $R^2 = 0.9293$ (PM2.5) and an F1-score of 0.9995 for toxic episode classification with zero missed spikes. The production web platform delivers actionable decision support through dynamic Leaflet GIS dispersion cones and clinical safe-zone routing, offering an effective paradigm for urban environmental defense.

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